2.6 Arithmetic expressions

expression

An expression is a combination of items, like variables, literals, ope
that evaluates to a value, like 2 * (x + 1).

literal

A literal is a specific value in code like 2.

operator

An operator is a symbol that performs a built-in calculation, like +,

Table 2.6.1: Arithmetic operators.

Arithmetic operator	Description
+	The ***addition*** operator is **+**, as in x + y.
-	The ***subtraction*** operator is **-**, as in x - y. Also, the - operator is for *r*
*	The ***multiplication*** operator is *****, as in x * y.
/	The ***division*** operator is ***/***, as in x / y.
**	The ***exponent*** operator is *********, as in x ** y (x to the power of y).

evaluates

An expression evaluates to a value, which replaces the expression
evaluates to 6, and y = x + 1 assigns y with 6.

precedence rules

An expression is evaluated using the order of standard mathemati
programming as precedence rules.

Table 2.6.2: Precedence rules for arithmetic operators.

Operator/Convention	Description	
( )	Items within parentheses are evaluated first.	In 2 * (x + 1), the result then multi
exponent **	** used for exponent is next.	In x**y * 3, x to tl the results then
unary -	- used for negation (unary minus) is next.	In 2 * -x, the -x is multiplied by 2.
*** / %**	Next to be evaluated are *, /, and %, having equal precedence.	(% is discussed
+ -	Finally come + and - with equal precedence.	In y = 3 + 2 * x, tl result then adde precedence thar y = 3+2 * x woul
left-to-right	If more than one operator of equal precedence could be evaluated, evaluation occurs left to right. Note: The ** operator is evaluated from right-to-left.	In y = x * 2 / 3, th result then divid

PARTICIPATION ACTIVITY 2.6.3: Evaluating expressions.

Start 2x speed

x = 4
w = 2
y = 3 * (x + 10 / w)

Preferr
y = 3 * (x + (

10 / 2
5

3 * (x + 5)
4 + 5
9

3 * 9

y = 27

Captions

1. An expression such as 3 * (x + 10 / w) evaluates to a value, using precedence rules. Items v
   comes before +, yielding 3 * (x + 5).
2. Evaluation finishes inside the parentheses: 3 * (x + 5) becomes 3 * 9.
3. Thus, the original expression evaluates to 3 * 9 or 27. That value replaces the expression. S
   27, so y is assigned with 27.
4. Many programmers prefer to use parentheses to make order of evaluation more clear whe

CHALLENGE ACTIVITY 2.6.1: Precedence rules for arithmetic operators.

763650.1443318.qx3zqy7

Start

c * d + 10

Which operator is evaluated first? Select one

1 2 3 4 5

Check Next

View solution (Instructors only)

CHALLENGE ACTIVITY 2.6.2: Calculate the values of the integer expressions.

763650.1443318.qx3zqy7

Start

Type the program's output

```
x = 7
y = x + 2

print(y)
```

9

1 2 3 4 5 6

Check Next

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